

BOMBER AIRCRAFT 1939–42

During World War II, two different types of bomber aircraft emerged: those designed to fly long-range “strategic” missions against infrastructure targets, or drop bombs onto industrial cities in the hope of causing significant disruption to manufacturing; and “tactical” bombers, which were employed on and around the battlefield. Both had vital roles to play, but it was the “heavies”, the four-engined aircraft capable of carrying many tons of bombs, which came to define the type. Aircraft such as these did not begin to enter service until war was already under way, operating alongside the twin-engined types, which had dominated production during the 1930s.



▲ VICKERS WELLINGTON

Date 1941 **Origin** UK
Wingspan 26.26m (86½ft)
Length 18.54m (61ft)
Top speed 410kph (255mph)
Engine Two 1,118.5kW (1,500hp) Bristol Hercules XI two-bank 14-cylinder radials

Metal-and-wood composite framework

Front machine-gun turret had power traverse

The “Wimpy”, as it was universally known, had a fabric-covered metal-and-wood airframe, which was light and very strong but took a long time to assemble. It remained in combat service with the UK’s Royal Air Force Bomber Command until October 1943.



▲ BOEING B-17 FLYING FORTRESS

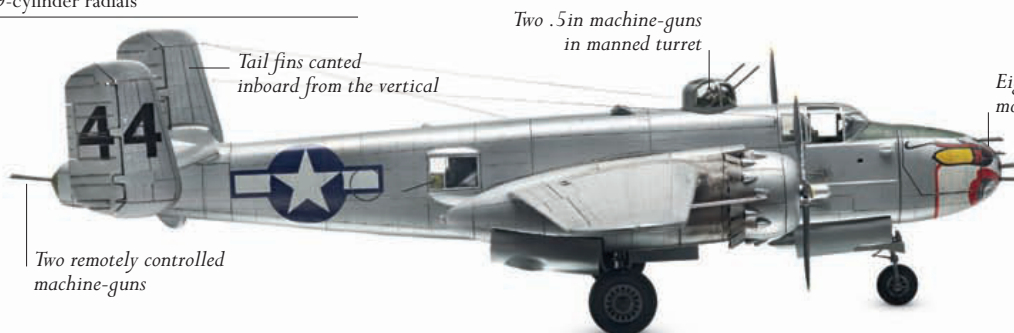
Date 1943 **Origin** US
Wingspan 31.62m (103¾ft)
Length 22.66m (74½ft)
Top speed 460kph (285mph)
Engine Four 894.8kW (1,200hp) Wright R-1820-97 Cyclone single-bank 9-cylinder radials

Enormous tail fin for high-altitude stability

Waist gunners had a single .5in machine-gun each

Two remotely controlled .5in machine-guns

Famed for its ability to absorb damage, the Flying Fortress’s (see pp.314–17) weakness was its limited bomb load – just half that of the Lancaster. Bristling with guns, Fortresses flew in “combat box” formation for mutual defence against fighters during daylight raids.



Two .5in machine-guns in manned turret

Tail fins canted inboard from the vertical

Two remotely controlled machine-guns

▲ B-25 MITCHELL

Date 1943 **Origin** US
Wingspan 20.6m (67½ft)
Length 16.45m (54ft)
Top speed 438kph (272mph)
Engine Two 1,267.6kW (1,700hp) Wright R-2600-92 Cyclone two-bank 14-cylinder radials

With war imminent, the US Army Air Corps ordered the B-25 Mitchell medium bomber “off the drawing board”, without waiting for a prototype. Eventually, almost 10,000 were built in six main variants, including the B-25H/J gunships, which had up to 18 machine-guns and a 75mm cannon.

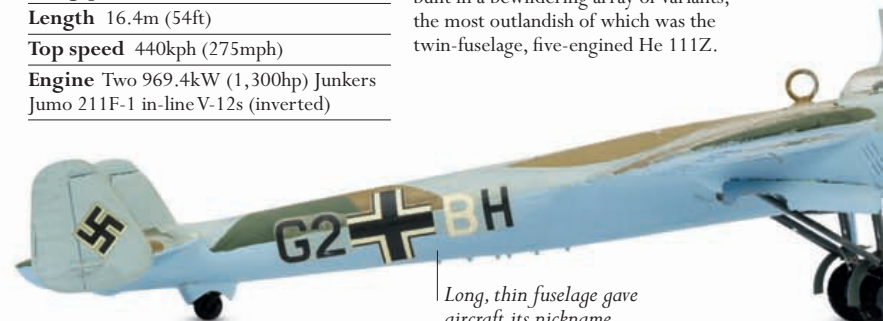


Ventral gun turret

▲ HEINKEL HE 111

Date 1941 **Origin** Germany
Wingspan 22.6m (74ft)
Length 16.4m (54ft)
Top speed 440kph (275mph)
Engine Two 969.4kW (1,300hp) Junkers Jumo 211F-1 in-line V-12s (inverted)

The most numerous of the medium bombers with which the *Luftwaffe* entered the war, the He 111 was built in a bewildering array of variants, the most outlandish of which was the twin-fuselage, five-engined He 111Z.

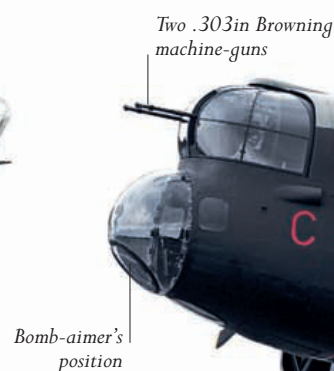


Long, thin fuselage gave aircraft its nickname

▲ DORNIER DO.17

Date 1938 **Origin** Germany
Wingspan 18m (59ft)
Length 15.79m (52ft)
Top speed 410kph (255mph)
Engine Twin 745.7kW (1,000hp) BMW Bramo 323P “Fafnir” single-bank 9-cylinder radials

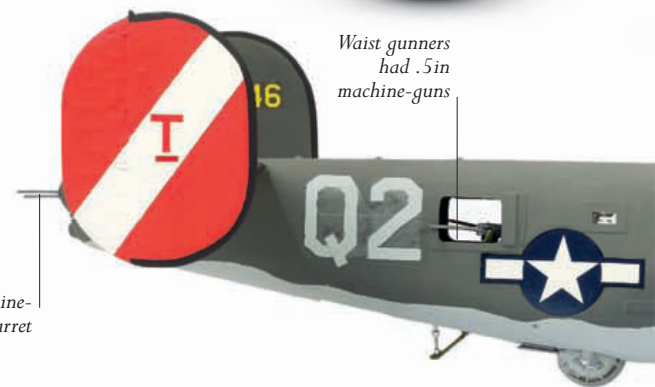
The Do.17 – the “Flying Pencil” – was relatively unimportant when compared to other German bombers of the early war years, especially the He 111. Thanks to its agility, however, it was a favourite with its crew.



Bomb-aimer's position

Two .303in Browning machine-guns

Eight .5in machine-guns mounted in nose



Waist gunners had .5in machine-guns

Two .5in machine-guns in rear turret